

ActInf ModelStream #001.4: "A Step-by-Step Tutorial on Active Inference"

ModelStream | 2:27:21

foreign

okay i think we're live okay yes the

screen was blocked out for a second but

hello everyone

welcome to the active inference lab this

is february

5th 2021 and we are here in

model stream 4.0 with ryan smith and

christopher white

i'm daniel friedman i'm a postdoctoral

researcher

in california and a participant in

active inference lab

maybe the two of you can just briefly

reintroduce yourself

um yes for anybody who hasn't you know

been watching the previous

sessions i'm ryan smith i'm a

investigator at the

laureate institute for brain research in

tulsa oklahoma

yeah hi i'm christopher white i'm a phd

student at the mrc cognition brain

sciences unit in cambridge england

awesome well we are here today in part

four

which will be the last section on this

paper hopefully not the last time we

get to speak together on any stream but

it's part four of four

on this paper a step-by-step tutorial on

active inference and its application to

empirical data you can find the link to

the most updated version of that

resource

in the video descriptions and as far as
points of process if you have any
questions or
comments or thoughts during the live
stream you can type it into the live
chat
and we'll address it when the time is
right if you have questions arising
after this live stream or you're
watching it after it's occurred
feel free to leave a comment and we'll
try to address it in future
sessions because the model stream never
ends so to learn and participate
and learn about how to participate go to
activeinference.org
that's where you'll find more
information on active inference lab
so today in our fourth session it's
going to be pretty exciting because
we've covered a lot of the groundwork
and
daniel are you crying ryan yeah you're
we're not hearing anything daniel
oh um
give me one sec okay do you hear me now
no yeah yeah you you froze
all right give me a second
it's all good this is how it happens
sometimes when
live streaming let me go back to the
event
and i will rejoin the event
all right
cool waiting this is just how it is
welcome to live streaming everyone
cool look at that ghost daniel friedman
he can be kicked out you're really quiet

again okay okay okay yes

device settings

okay yes thanks microsoft

teams for facilitating these

conversations

i it's still live and happening it's

just a little

a little hiccup with teams that now

we've gotten that out of the way

so it's fixed but thanks ryan and i

don't know if this other person can be

kicked out

but take it away we're all good though

thank you

okay uh maybe um

which one are you the ghost of daniel's

past

it's okay it will auto drop them um

like after you share and unshare but

it's all chill okay okay

perfect okay

i think that that worked okay so we are

still

we are still alive and going yep okay

all right so uh so i should just start

or is there anything else you were kind

of in the middle of saying something

just um if we could start with some

recap for those who probably have seen

the first three

sessions just where did we get to as of

today and then where does that take us

into the future

um okay so yeah just as a quick

recap um as daniel said the the kind of

whole purpose

of this tutorial that we

put out was to try to make methods for

actually building active inference
models
and applying them to experimental data
we wanted to try to make those methods
more accessible to a broader audience
um because up until now it's been fairly
difficult
to find sort of clear resources for
beginners
to start learning these sorts of methods
and so
the whole kind of the whole you know
point of what we've covered so far has
kind of been building to this section
um if the focus again is on b
being able to actually use these models
for experiments
so obviously the earlier sections could
be useful on their own if you want to do
simulation work and there's lots of
papers
reporting simulation work but in terms
of
then applying the models and simulation
work that you do
to um to actual experiments to actually
fit it to
um task behavior with participants and
you know
predict things about like neural
responses and fmri studies and all that
kind of stuff
then it all kind of boils down to what
we'll talk about today
um so in earlier sections we kind of
covered the basics of active inference
you know what free energy
what variational free energy is what

expected free energy is
what the motivation is for doing active
inference what the
potential benefits are over
traditional reinforcement learning
models um you know things
things like things like that just kind
of the nuts and bolts and then we
covered how to
build um active inference partially
observable markov decision process
models um and um as an example
um we built a um a simple kind of
explore exploit model
which i will uh or exploit what task
model should i say
as a just kind of a concrete example of
how you would build a model
um and we showed some simulations both
for
perception and decision making and for
learning um
with that model um and so
and then we also kind of um uh
took a took a break from that we're
going to return to that now but we also
took a break from that a little bit
to um cover um hierarchical models
and the neural process theory um
but now we're going to come back to
using the explore exploit task model
because it's
um simpler and more straightforward in
terms of
what you do with participant behavior
and and how you fit it
um so so all that like i said is
building up to

using the task model we built as an
example for
fitting it to data um so christopher i
don't know
yeah great summary christopher what
would be your perspective or what was
something that stood out to you
from the previous sessions yep because
you're
you're so both deep in the game and
actually producing
this artifact that it's been fun for us
and for the audience just to be
kind of talking through it because it so
what what does it
leave upon somebody who's been involved
in the generation of this artifact
now to be talking about
interesting question i suppose um
i thought i think it has made and i'm
just so this is kind of in
for context i have i started my phd four
months ago
so this is all kind of in the context of
figuring out
not just how to build these models kind
of for myself
um or sorry how to teach other people
how to build these models rather
but actually how to build these models
for myself and apply my own research
questions
um and so i actually have never fitted
them to data
all of the stuff that i've done so far
has been theory
essentially just i've done a lot of
simulation work but no um

no data fitting so
i don't know i think that's actually the
the frontier in active inference at the
moment is actually figuring out how to
get these models to hit the road
um and there's a really one one really
tricky thing is most the people who work
on active inference
in a real way are neuro images
so i for i for example i'm in your
imaging department
um i know ryan you work in your imaging
as well
um it's actually challenging to test
these models in the domain of
neuroimaging
neuroimaging is really hard um actually
i think many of the clearest predictions
actually in terms of neural process
theory are actually things that you
could test
really quite easily with optogenetics
not optogenetics or anything like that's
easy
but say predictions about micro
circuitry in pfc
which active inference certainly makes
um
can very much be tested easily in
a rodent model but it's really difficult
to test it in
a human and so i think
the point that i'm trying to in a rather
rambling way get to
is i think we need to think very deeply
about when we build models
about what it actually is that we're
trying to solve do we just want a

directional hypothesis
in which case you don't necessarily need
to fit these things to data
you're just saying a's b a will be
bigger than b which is still better than
most psychology
let's let's be honest um
yeah it's it's really interesting the
neuroimaging case
has a few specific difficulties massive
amounts of data
in a system that has massively complex
internal
and external dynamics like the brain and
you're making a ton of data with a very
complex error profile and in the spm
textbook
it's chapters and chapters of
normalization and re-warping and all
these different kinds of ways to deal
with these very complex data sets
and now we've almost built out that
framework and now just like you said
where it's hitting the road
is in being applied to empirical data
and it's a little bit of a surprise
at least to me and probably others that
maybe neuro imaging although that's the
home
of spm we're seeing it applies to
systems beyond
neuroimaging and even there's other
systems that help inform
our neuro imaging quest so pretty
interesting stuff
um okay so should we just get started
then sounds good and anyone in the live
chat just

drop a question and we'll get to them

thanks a lot

okay so one brief thing

that i did want to mention um is that a couple times ago

we um we talked about we covered learning um

but since then um just wanted to let people know we actually caught a little a small error in the learning section

um in the previous version that we've now corrected and it has to do with the way that you

calculate the um the novelty term and expected free energy that gets added to um it gets added to expected free energy

when you're doing learning so it's what drives the parameter exploration

um term expected free energy it basically drives an

agent to seek out policies that will tell it more about in this case what the a matrix looks like

so tell it more about what the relationship is between states and observations

and there's equivalent terms that could be added if you're trying to learn transition beliefs or

other matrices and vectors in the model but um

so now just to kind of make that clear we actually have added two different additional worked examples

of how um of how to calculate the uh the novelty term just to make sure that's clear

so um if if anybody doesn't happen to
have the kind of most updated version
and you're interested in kind of knowing
the kind of rigorous details
about how learning works um in the uh
in the formalism then um you know i it
might be helpful
um to download the updated version so
that you can
have additional work examples to to get
a better sense of how that works
um so i just wanted to um make that
uh clear like i said it wasn't it's not
like a big error but it's a small error
that if you
were to read it it could be confusing um
so
i just wanted to make well just wanted
to highlight that for people
um but today like i mentioned
the focus is going to be on actually
building task models
so and applying them to data and there's
actually
you know it's probably you know this
doesn't this isn't necessarily like
the only way you could break it up but
i've broken this up into kind of six
steps
that um are probably a good kind of
heuristic
um you know road map or or whatever you
want to call it
for doing this kind of work so the first
thing
is that you have to have a task and you
have to have participants perform that
task

the second thing is you have to build
one or more models of that task so
one or more models in this case
partially observable markov decision
process models
in the active inference formulation
that can reproduce or generate u^m
simulated behavior for that task
 u^m and so once you do that
then the next thing you need to do is
you need to find somehow find parameter
values
in each of those models that you
construct construct
that can best reproduce each
participant's behavior
so for instance under one model you
might find that
the first participant the model can
reproduce the first participant's
behavior the best
if it has a particular u^m
say value for how much they want the
reward so for instance like the value of
the u^m
or the think about it as a precision of
the preference distribution
or you might find that for another
participant
you need a lower value for that to
reproduce their behavior well
and but they need something like a
higher learning rate
or a lower action precision value
right there's all these different
parameters in these models we've been
covering
and different values for those

parameters are going to be better different parameter value combinations and these models can be better at reproducing one person's behavior versus another on the task um and so what that requires is having some kind of estimation algorithm um something that basically searches through in some way different combinations of parameters and checks how well it reproduces um a given participant's behavior um and so and so like i said those are called parameter estimation algorithms and there's a few there's several different um types that you can use um i'll briefly mention a few and then we'll go into detail on the one that is probably most consistent with um the general kind of theme of active inference models which is a scheme called variational base so once you've found for each model the parameter values that best describe each participant then you want to know okay well which of those models is the best model and so in that case you need some kind of way of comparing how well um each model on average can reproduce the behavior of the whole group of your participants so each model is going to have their best fit parameters for each participant but still the best fit parameters for

one model
might still do a better job than the
best fit parameters for another model
and again we'll go into this
and then once you've identified what the
the best model is
then and this is something that might
not be obvious but it's really important
is you have to somehow confirm that the
parameters in the best model
are recoverable so you can either talk
about this as whether the
the model is identifiable or that the
parameters in the model are
recoverable and that's something that
again i'll expand on
but the but the basic idea is more one
one kind of simple way to think about it
is um what if it's the case
that two different combinations of
parameter values
are equally good at reproducing or
explaining
a given participant's behavior if that
were the case then there isn't any
unique
combination of parameter values that is
kind of
the best set of parameter values
and what that means is that
you could take one set of parameter
values
generate behavior with it but when you
ran estimation on that
you could get a very different set of
parameter estimates
and if that's the case then the
parameter estimates that you're actually

getting for that model for a participant
aren't necessarily a reliable or
uniquely best
description and so that means that
parameters aren't recoverable or there's
no kind of
uniquely best set of parameters under
that model
so you need to do that to make sure
that's true for the winning model
and if it's not then you might move down
to say the second best model and see if
that's the case
right so you want to find the best
fitting model ideally
you want to find the the best fitting
model that is also
clearly recoverable that has clearly
recoverable parameters
finally once you have identified
the winning recoverable model um
you have this set of parameter estimates
that describe each person so these are
individual difference variables right
one person has higher action precision
than another one person
has a more precise preference
distribution than another
and at this point you can take these up
to the group level and you can say okay
you know for instance does a healthy
group
have different parameter values on
average than
say a group with depression or a group
with anxiety
or you can say you know do parameter
values predict something on a continuous

you know in a continuous way right maybe
higher anxiety levels are associated
with
lower action precision
or something like that and so i'll show
you guys examples of this kind of thing
as well
um so first and you know we've kind of
already covered this in previous
sessions
or just examples of it but there are
lots of different kinds of tasks that
you can
model which is kind of a nice perk of
active inference models
is that they're not just about decision
making as a main component you can also
use them to model
kind of simple perceptual tasks so
christopher in a previous session
showed how you could build a model of a
simple perceptual oddball or local
global
task which is almost entirely about
perception
you won't get necessarily interesting
individual differences in behavior in
these tasks
but you can show differences in say
simulated event related potentials and
eeg studies
and in principle you could fit the erps
to find the best parameters for a person
but you can also do um inferential or
inferential or prospective decision
making tasks like where you
plan for the future and make a decision
based on what you expect future outcomes

to be
um which is the kind of thing um uh
kind of like what i'll be showing you um
or with the explore exploit task um but
the explore exploit task
that we'll be showing is is kind of very
similar to a standard kind of
reinforcement learning task it just
requires that you
seek out information before you can
really know or even learn
what the right reward values are
and then lastly you could combine this
with neuroimaging
in the context of predictions from the
neural process theory which is what
we've talked about
and what we covered in relation to the
hierarchical model that christopher
covered
so i should say that this is kind of a
fairly new emerging
field and so there's not that many
papers um empirical papers that do this
kind of thing
um you know it's it's something that my
lab has kind of been trying to
make more common uh practice
which is again part of the motivation
for this tutorial is so that
a larger number of researchers can do
this kind of thing
um but so you know i mainly work in
computational psychiatry
um and so you know we've this has been
used to show say for instance
less uh less precise um lower action
precision

and substance use disorder and also um
slower learning rates for negative
outcomes and substance use disorder
you know we've shown things like that
greater decision uncertainty
is associated with uh people with
depression anxiety and substance use in
the context of
approach avoidance conflict tasks
most recently we also used kind of a
simpler
bayesian perception version of an active
inference pmdp to show
differences in the sensory precision in
an interceptive context so for instance
when people
how how precisely or accurately people
perceive their own heartbeats
and interesting differences there
between
again clinical multiple clinical groups
and healthy participants
and then lastly and you know one of the
first papers using this kind of approach
was actually philip fortenbeck's
paper where they looked at
predictions about mid-brain mid-range
dopaminergic responses um
and uh changes in the expected free
energy precision
term that we had talked about um in
previous sessions
and was able to show nice relationships
between
these expected precision updates and um
and uh what and neural responses using
fmri
in a region of the midbrain that is

known to
be rich in dopamine neurons um
so this is just kind of these are recent
examples um
that you know we're hoping other
researchers can can you know
build on um on that neuro slide
these examples we're seeing a couple
examples of different technologies
like fmri but we also heard it goes
beyond neuroimaging the scope of these
methods
and we're also seeing a couple of
biological questions or
conditions um related to your clinical
experience
let me complement those two axes of
variation with the methods in the
biological question
with this algorithmic dimension and
there's a question in the chat that says
what kinds of tasks
reflect the relationship between dynamic
programming and active inference
for example bellman optimal state action
policies thanks for the elaboration
so how do we connect some of these
biological and methodological
uses and axis of variation to some of
the algorithmic questions about
optimization dynamic programming so
um yes i mean it's a great question we
actually have a recent um
pre-print that uh lance decosta's first
author on
um um where he was able to come up with
some proofs to show when active
inference

is and is not bellman optimal um and
in terms of what the the conclusion was
in terms of
one-step policies um active inference
is bellman optimal in the context where
um
the uh precision of the preference
distribution is maximal uh
it's maximally precise over whatever the
reward outcome is
um active inference models in the kind
of simple md
pomdp scheme we've been talking about
here um they're not bellman optimal
or they're not guaranteed to be um in
the context of uh multi-step so like
deep
temporal policies um but um
whereas the um more recent uh
sophisticated active inference
algorithms
um are uh bellman optimal um for deep
policies
uh that's because they more or less
correspond to backward induction
um so um that's that's um
the probably shortest answer i could
give to that
christopher did you have something to
add there
uh no i yeah i was just going to point
to the lane slot to costa paper
great i posted the cost of papers that i
think are relevant in the chat so
thanks for that question and continue
okay
so just to be clear this is a very
specific paper

that you know lance and i and nora and a
few other people put out as a preprint
just a couple of months ago um so so i'm
talking about one particular paper
of lance's um um so i mean i can
certainly that point that
that specific one out so people know
which one i'm talking about
the relationship between dynamic
programming and active inference the
discrete finite horizon case
with yeah that's the one decosta sajid
par fristen and smith
yep okay cool um
all right so hopefully that helps um
um okay so now i'll kind of be going
through um each of the the steps here
that i was talking about
um kind of one by one right with our
explore exploit test
example so first step one is you have to
have participants perform a task
right so in this case um and i should
say for some of this it's really going
to be assumed
that people were following along with
previous sessions
um the time constraints just really
don't allow us to cover
all the background um so that somebody
could watch this one without having seen
the other ones and fully follow
everything i'll be talking about
um so just just to make that clear if
anything is kind of confusing then the
previous sessions are kind of necessary
we walked through the matrices and some
of the degrees of freedom

in this setup so i think it should be
good check out the other videos if you
haven't already
but but you know as a brief you know
kind of refresher about the task that
we're talking about
um the on each trial the agent starts in
a start state
and then he can either directly choose
one of two slot machines
and if they're right they'll win four
dollars but if they're wrong they'll win
zero
um and they start out not knowing you
know
like just flat prior over whether the
left or the right one is
whether it's the context in which the
left one or the right one is more likely
to win
so it's just 50 50. so it's pretty risky
to just pick one right away
at least before you've learned if one of
them is actually more likely
if the context in which one is more
likely to win is more common
so instead the other thing that the
agent can do is they can choose to get a
hint
and the hint will tell them which um
whether it's a context in which the left
bandit is the one that
is more likely to win and when it's the
right one
um and in each of the contexts in the
left
in the context where the left one's
better it will pay out eighty percent of

the time

uh whereas in the context of the right one uh the right one will pay out eighty percent of the time

um but the catch is if you choose to take the hint then you'll only win two dollars if you get the right one afterwards

um so this trades off this kind of reward seeking and information seeking where if you seek information first you're more likely to win but if you choose first right away then you'll get a higher reward if you're right

um so it specifically trades off reward and uncertainty

um and this is where a lot of this is not going to make sense unless you've seen the previous sessions

um is that and the second thing is you have to build one or more models of the task we just talked about in our example we'll use two different variants of the model

one where we're just going to fit the risk seeking parameter

and an action precision parameter and a second model where we also assume that the agent has a learning rate or that there are differences in learning right

so for those that don't remember um the risk seeking parameter just corresponds to the magnitude of this α value

um so what this says is at time two if the agent observes a win so this row

is a win then
the value of that win will be r
and if they instead wait and choose the
hint they'll continue to observe this
start thing
at time two but at time three if they
win
then the value they will win will be
this r value divided by two
um so so whatever r ends up being
the win at the third time point if they
take the hint is half of that
um so we're going to be fitting for each
person what that r value is
which again stands for risk seeking
which would be clear to anyone who's
watched the
previous sessions because the higher
this value is
the um the less exploratory drive the
agent will have
and that means that they'll be more
likely to take the risk and just
choose one of the slot machines randomly
as opposed to
taking the hint first to be more
confident in which one's right
so the second parameter is this action
precision parameter
 α and all this basically says
is the precision or the probability of
selecting some action
given an α value corresponds to this
um so our corresponds to
whatever the probability is of an action
given the
given a policy μ scaled
by this α thing and then softmaxed

to become
a um just a proper probability
distribution again
um so it just controls how more or less
it controls how precise action selection
is
given the choice of policy
and then finally in the model in the
second model where we're also going to
assume an agent has a learning rate
um other than just the optimal learning
rate of one
then we're also going to be fitting this
thing η here which is the learning
rate
and this just says that your beliefs
about the probability of the context
being
that the left slot machine is better
versus that the right slot machine is
better
just updates based on your beliefs about
whatever the that state whatever the
winning state was
on the previous trial again four times
step one
so basically each trial whatever the
posterior beliefs were
over states um they
the count for that just gets added to
whatever the prior
initial states was um athena is at the
next time point
but it's scaled by this η thing so
you know if η is one then it'll
attack on a count of one
again if it knows with certainty that
it was in say like state one

but if that is 0.7 that only add a count
of 0.7

on each trial given that the posterior
distribution was like one

zero um um so

so that's um the the basic setup is
we're going to fit either just β and
 α or we're going to fit both of
those and η

um one note on that ryan i like this
step

to build one or more models because
we're thinking about

active inference as like this process
theory it's instrumental

you know the whole instrumentalism
versus realism question we're using it
as an instrument

and that means that we're iterating over
models we're relating our models to each
other in specific ways like make the
simpler one and then make one that just
adds one more variable

and then it's not like you're asking
what is the active inference model here
it's here's the active inference
approach

and we're going to combine our
perspectives and make multiple models
and we'll be comparing models so the
only limiting factor there is

how many models can we think of at the
beginning and how wise can our model
selection process be

so it's just really like a pluralistic
but pragmatic way to go about it
by saying construct one or more models
because then you're never going to fall

in the trap of thinking you're making
the active inference model because there
isn't the model of anything
statistically
they're just instruments so it's a
really helpful perspective and so this
isn't just a little
um convenience to make one or more
models it's something that reminds the
scientist
or the modeler that they're actually
making one of many or infinite possible
ways to think about a situation
yeah i mean i mean hopefully you've
really kind of you know thought deeply
about what
the most plausible models are given
current theory and you know current
previous or previous empirical research
right so i mean they should be informed
um but yes so you're just you're trying
to find
the model that has the highest evidence
that the observations or the behavior
provides the highest evidence for
and yes it is always possible that
there's some other model
that you haven't thought of that would
be better um but
you know if you have a model that
explains behavior really well
and it does so better than any of the
other ones you can think of then it's
kind of a good
bet um as at least giving you um
interesting individual differences
to to again bring up to a group level
analysis

um so so okay
so you know to get kind of a little
nitty gritty in terms of the the
code and the matlab structure and
everything um you know the kind of thing
you need to do
is you first so you have a participant's
behavior right like
they chose the hint and then slot
machine one that shows the hint and then
slot machine two okay now they just
chose the whole
machine right away over slot machine two
right away
um and so forth and those have to be
coded in
right to each trial in the mvp structure
which
which again uh previous sessions will
explain
um so the mdp structure in matlab
um for each so each uh
each number right inside these
parentheses here will
be the trial that you're talking about
and the dot u here
is the um the matrix that described that
encodes the actions that a participant
took um
now if we remember um so row one here
corresponds to state factor one which is
the context
whether left or right is better and
there is no action for that that's not
something agent control it's just stable
within a trial which one it is
so these just get ones and there is no
other possible action

for the second row though so this is the state factor that corresponded to action selection um two you know action two corresponded to taking the hint an action three corresponded to choosing left so if that was what a participant did on trial one then you would have to feed in two followed by three right they chose the hint and then they chose left on that trial um and so you just have to iterate that over all the trials in the task so one to you know end trials um whatever they did um so that's mdp and in this case like i said there are two actions so two columns and there are two state factors where only the rows which only the second one is control um so now mdp o.o is the observations um so again if you're familiar with the um the way we set up the model for this task before there are three different uh outcome modalities three different types of observations you can get so the first um observation modality what a hint or not so in this case one is just you're kind of still observing the start observation if you observe two that means you get a hint that the left one is better and if you observe a three that's the hint that the right one is better

so in this case they got the observation
that the left one was better when they
took
one two one because after they got the
hint they just returned back to
observing kind of the initial
observation
for the hint modality um whereas
um for the second modality here which is
the wind's losses
for the first two time steps they just
observed kind of the starting
observation here
because they chose the hint at the
second time point so there was no win or
loss
and then at time point three they
observed a win so
win is encoded as a three in this model
and two or losses is encoded by a two
um and then uh the final
outcome modality is just the agent
observing what it does
so in this case this just says started
in start state
chose to take the hint and then chose
the left uh
chose to it observed itself choosing
the choice of the left slot machine
so that's it so all you have to do when
you're
instead of simulating behavior you're
actually um
fitting that way you're actually feeding
in participant behavior
to uh fit it uh to fit a model to it um
then you just have to you know sort of
translate the behavior and the raw data

into a form like this that matches the action and outcome representations in the model um and um and so that's it you first feed those in um and then once you've done that then you have to use some kind of estimation algorithm which basically means the thing is going to somehow repeat simulating behavior in that model until it finds a model that maximizes the overlap between the probability of choosing the participant's action of the probability of choosing actions and what the uh participant actually chose right so for instance if it shows hint and then left bandit left slot machine on trial one um it's gonna find a set of parameters that apply across all trials that maximize the probability that the simulated agent also would choose the hint followed by choosing the left slot machine so there are a number of different estimation algorithms that can be used um you know the the simplest possible one is just something like a grid search so in this case say you just have two parameters which this is just arbitrary called alpha and beta you can kind of divide this up into little grids different combinations of parameter values so for instance like

you know parameter 0.2 for alpha
and parameter value 3 for beta
and then you can just encode for example
what the probability
of the model is given the participants
actions
um or um that would be the posterior
which i'll talk about or just the
probability of participant behavior
given the model and those parameter
values i mean so in this case this
is kind of clear nice result where this
value right here
something like point three and four
that um is maximally and uniquely
the best set of parameters for
reproducing that participant's behavior
and and in this case you're using
something called in this case you're
using something called maximum
likelihood
so maximum likelihood estimation
which just means again you're trying to
find the
model and set of parameters for which
the
uh probability of the participants
behavior given that model is highest
which is a type of likelihood right so
you're trying to find the maximum value
for the likelihood
which you just in this case encode with
these you know redder colors
equals higher uh likelihood
of behavior given the model you can also
do something
a little bit um so to be clear what we
actually would want

right is there is the inverse of this we would want the probability of a model given participant behavior which is the posterior but when you're doing maximum likelihood estimation you're just assuming here that you have essentially a flat prior for what the probability over models is so in this if that's the case then the probability of the model given participant behavior is just proportional to the likelihood but another thing you can do is called maximum posterior or map estimates which is the same thing except it doesn't assume that you have a flat prior um overalls right so if you just start out with a prior expectation that one model is better than another then you can incorporate that information into um into parameter estimation and this is kind of an example of this where um you might find this distribution in terms of just the likelihood right the probability of data given the model where a parameter combination right around here would be best but then you might also have some reason to have a prior expectation that models down here right are actually more likely and so if you combine those guys

together you can get a posterior that looks a little different right like a something very similar um a very similar model wins in this case but that's just because the behavior that's driving the likelihood here is just already a really good fit so it kind of dominates over the influence of priors but that's not always going to be the case so so those are two approaches um and and again you can you can do them with a simple grid search or you can do something so i should say that the grid search kind of thing it only really works when you have a few parameters we don't have very many and the reason is because the more parameters that you have the higher uh dimensional um this uh parameter space becomes and it just becomes it it takes sort of intractably long um to do to search through every possible combination um so what you do instead is you use some kind of gradient descent process um exactly like the gradient descent on free energy that we've been talking about for how active inference models arrive at posteriors over states and posteriors over policies um so there's a really kind of interesting overlap between the way that you estimate models

via gradient descent and the way that you actually the way that active inference models update their beliefs via gradient descent um so in this case or is a very good point and i just want to point out one similarity and difference because people may have heard about these gradient descent algorithms for descriptive statistics or for fitting descriptive models so that's from going again from the data that's empirical to a descriptive model or descriptive statistics like regression coefficients and if it's a multi-variable regression you might need to use not just a grid search but act that you can do a simple maximum likelihood approach like a least squares approach or something like that you need to use this kind of a complex multi-dimensional optimization to get to that descriptive statistic and then there's this little twist where actually in active inference we might be using those computational techniques but we're estimating the parameters of an underlying generative model and then there's this nice little return where actually the generative model is implementing that as well yeah so so in so in this case you you absolutely need some kind of prior value because with gradient

descent you're not exploring every combination what you're doing is you're starting at some starting point which is just coded by a here um and then you're just like when you're minimizing free energy in in an active inference model you're searching neighboring values and trying to find a value that has a higher likelihood or a lower free energy and then you just kind of keep doing that iteratively until you find some value that where the likelihood stops getting bigger or free energy stops getting smaller and that then corresponds to your best estimate or your posterior posterior estimate for the parameters for a given participant and so when you're doing this with when you're doing it with a gradient descent for uh free energy then um that ends up corresponding well to um the actual approach that we'll be talking about um or actually using in our example which is a variational base so the variational bays you do exactly this you set a prior mean value and a prior variance for each parameter and then you do gradient descent until you find a set of parameters that minimize free energy

which is again an approximation to the model evidence

so but but it's important to keep in mind some potential limitations of this approach which is if you see here this parameter space it does look like there is a kind of nice single right like local minimum right the local place where the free energy is lowest

or um where the uh again it would be a local maximum be like top of a mountain in this if you were talking about the highest

likelihood um they typically do log likelihoods as opposed to likelihoods

and but one of the issues is this sort of

the parameter space need not have a landscape that's quite this clean you might end up for instance having a landscape just moving to a two-dimensional case now

that looks kind of like this where if you start out say with a prior value with prior values that um you know like this kind of red

circle up here then if you do gradient descent

you might end up getting stuck in a little local minimum like this

um where you know and then and then the gradient descent algorithm would say hey you know actually

you know this seems like the best one because if i move in any direction then the free energy goes up um

whereas actually the global minimum the one that you would want to get to is this different one that's on the other side of this little kind of free energy hill um and so you know this is this is an example where i'm choosing the the right priors is important um to get the to be able to get the the best parameter estimates um but also this speaks to the kind of thing about parameter recovery that i was talking about earlier because it could be that if you set your parameters kind of right on the top of this hill thing here then even if the parameter combination over here is the one that actually generated the data you could get stuck over in this one in which case the estimation algorithm would not give you the right parameter estimates so this is why what i mentioned earlier about assessing parameter recoverability is really important so all my way of saying it's important to choose good priors or to figure out what good priors are um and there are there are interesting ways to try to optimize that so there are kind of hierarchical bayesian techniques we won't talk about these explicitly but where you can kind of and these won't necessarily solve this problem but where

you can

more or less you can think about it as
it estimates the parameters for each
person

and then it notices hey you know looks
like these are all shifted over
you know the distribution of these looks
like they're all shifted over say to the
right

you know so actually the looks like the
distribution is more kind of over here
and so then it might um choose that
um choose that uh that the kind of mean
value of that

um as the new uh priors

um for uh for redoing the estimates

um until it finds uh values that um

that essentially it's it's helping you

to find whatever the optimal priors

would be

um but uh but again that won't um that
won't always

solve this sort of problem with a lumpy
landscape

you know um one other little thought

it's actually almost three

layers with the red dot so there's us as
agents on a landscape

we actually need to come up with policy
as agents in our niche to get around

local minima like my door's closed i

need to step away to open the door

before i can move

through it then how are we going to fit
policies for ourself

under pervasive uncertainty well we're
doing this state estimation with the
policy

so that we can actually come and then
how are we going to converge on those
parameters
using a a little bit like a gradient
descent
algorithm internally not internally to
us is just within the agent
computationally it's just it's very
interesting because people may have
i'd imagine there's one kind of person
who sees this and says oh we talked
about optimization for a year
in my course so i've heard all about
non-convex non-complex optimization
techniques or rugged fitness landscapes
and there's another group of people for
whom actually this optimization theory
might be quite novel because of
how they've looked at modeling before so
very interesting
and to put into the fourth section like
that really spoke a lot
so interesting stuff ryan i just want to
ask christopher you want to add anything
before we continue with this
no this really is very much ryan's part
of the paper oh cool
yeah yeah i'm just here kind of oh if
anything
more than welcome to add thoughts or if
you have any other insights about
technical aspects
no no nothing really um i would just
i might ask some questions if something
if things come up
um because i'm gonna be doing this at
some point in the next couple of months
but um so yeah

great thanks okay so
um so um like i said the the kind of
detailed example that we are going to
use is variational days
um and note that variational message
passing is
again what agents are using within
um within active inference models
which we talked about before so we're
using something very very similar to
this
or i mean so technically i should say
now it's using marginal message passing
in the latest
versions but again we cover this it's
very similar um to variational message
passing it's just kind of a slight
improvement
um but so we're using um the same sort
of approach
to estimate parameters uh to estimate
the parameters that people are using
uh the parameters that are kind of you
know potentially kind of uh stored in
their brain
in some sense um so
so we're doing gradient descent on
variational free energy as i mentioned
and as i mentioned you need to specify
prior means and prior variances for each
parameter
and then you just move the prior values
in the direction
of increasing action probabilities um
but what you're doing because it's a
gradient descent and variational free
energy
is you're not technically just trying to

find

the maximum likelihood value like you're
doing with like a grid search

um instead um what you're trying to do
is you're trying to

well i should say that the variational
free energy part of it

means that there's a complexity penalty
um and what that basically means is so
if people remember from

variational free energy when we talked
about it before the simplest way or kind
of intuitive way to think about it

is that it's just complexity minus
accuracy

right and what complexity means is how
much you have to change your beliefs
so if you start out with particular
prior values

then the farther you have to move those
move your beliefs from those prior
values

the that's going to push a variation for
energy up

whereas at the same time free
variational free energy is going to go
down

as the model predicts behavior better
um so what it means is for instance
you know if i start out with a prior
value that's say way down here
like around like i don't know one and
three

then it's going to have to move from
those prior values

a pretty long way before it gets to
a set of parameter values that fit well
whereas say if i started over here it's

not going to have to move those as far
but if i were to do that then
potentially
instead of the posterior estimate
posterior parameter estimates actually
settling on this thing that has the
maximum likelihood
it might stop say like around this one
right a little lower than it or
something like that
um because that maximizes that leads to
high
accuracy while also not having to move
the parameter values
uh really really far not having to
change beliefs too much
from what the um uh informed
prior belief was um so it's you know
it's
it's important right to think that this
assumes something that
that the priors you have that you have
them for a reason right that they're
actually based on something
that they're informing you um so that
so that it is actually does actually
make sense not to move them
too far from prior values and in
practice the reason this is helpful
is because one thing that often happens
when you're just doing maximum
likelihood
is that um that maximum likelihood will
be with respect
to your particular data set right your
particular
set of parameters or set of participants
um but often if you choose just the

maximum likelihood value
then that's kind of over fit to just
those participants
where if you were to say apply that
exact same model to a new set of
participants
um it might not do as well because
it was fitting you know specific things
that were not generalizable about your
data set in particular
um so by by um
by putting this kind of complexity cost
on it it prevents overfitting which
means that
the predictions of that model are more
likely to uh generalize
to a new set of participants later so it
prevents overfitting which is a very
common issue in in just standard
frequency statistics as well um
um okay so so that's so that's what
you're doing with variational base is
doing this kind of
complexity minus accuracy thing where
you're preventing overfitting while also
maximizing
the accuracy of model predictions um
okay so then like i mentioned you need
to do model comparison to identify what
the
best model is um
now uh uh let's see so i just realized i
should probably set something in motion
here let's
see if i need to uh so when we actually
go into the code here in a minute one of
these things takes a long time
um let's see so i set it to store stuff

i just want to triple check that i
shouldn't um you know set this thing
um let's see
uh yeah i bet i should have double
checked this beforehand
um

[Music]

cool though it's really
interesting and christopher you want to
add anything or maybe just a quick note
while ryan's figuring out like what is
it like to be learning these models or
what kind of skills do you wish you had
earlier when you were learning the
models um
other than of course your own tutorial i
mean

[Laughter]

i think everyone who did undergrad
psychology is probably
at some point in there i i did cognitive
science as an undergrad but
um i took a lot of psychology classes
when you take staff psychology a lot of
people read this andy field textbook
and he has a chapter on um
uh some type of like esoteric regression
and introductory sentence is like i've
never done this
i don't see myself ever doing it but i
wrote this chapter about it and if i
ever need to do it i will be very
impressed by how much i seem to know
um so that made that probably will
describe my experience with um
this tutorial to a certain extent uh
okay so yeah so real quick i'll just i'm
just gonna

jump to the code and explain why i
should have started this thing
is that that so what if you set this
thing to sim equals 5
then what it's going to do is it's going
to actually generate simulated behavior
for six hypothetical participants
where each participant has a different
combination of parameter values that's
generating
that simulated behavior and then what
it's going to do
is it's going to then apply the
estimation algorithm
to the resulting simulated behavior
and it's going to give you a set of
results
and then what it will do is it's going
to do bayesian model comparison on those
to identify what the best model um
is and then i'm going to try to do this
i'm going to try to set the so i'll just
say for
for um for
the purposes of um you know you actually
you guys actually doing it
yourself at home um
i'm going to uh
i've set this thing to 32 trials for
each
uh simult for each set of simulated
behavior
but maybe if i set this to 16
then it will go faster just to think
about what's happening
people looking through these equations
every time two matrices or
two numbers are getting multiplied the

computer has to do something
so we're kind of nesting matrix
multiplications
inside of even bigger ones so if you
want two time steps for two
larger time clicks for four models you
know for four participants it really
starts to balloon
rapidly especially when there's
computationally intensive steps
yeah so here just uh just to show you
guys what's going on so i just started
this thing
and if you look in the matlab window
here
it's calculating log likelihood so ll
stands for log likelihood
over and over again under a particular
set of parameter values
and in this case it's negative 31 and
it's trying to
minimize that or in this case it's
moving it closer to zero so
maximizing in a sense but bringing it
closer to zero so now it moved via
gradient descent to a second set
of parameter values and now it's found
that okay now the log likelihood is
negative 22. so we're even closer to
zero
and again same thing now we're at
negative 19 and it's going to keep going
until it settles on a stable value
and in this graph that will update
every time which i'll uh
show you here so this is just showing
ignore i should say these so something
to explain

is these routines were originally
designed for dcm for dynamic causal
modeling with fmri
so um a lot of the a bunch of these
graphs and also the the sorts of labels
that they have
are don't really apply in this case they
only apply to dcm
um so you know so ignore a lot of the
labels but the point here is is that
each iteration as you go from left to
right is a new estimate
of the log likelihood for
every um for the set of parameter values
it's trying
during gradient descent um and so you'll
see
that after a while the thing is just
going to kind of
plane out it'll converge onto a value
you see right now is it's
like 17.35 and now it's like 17.16 so
the thing is kind of
starting to converge on a stable minimum
log likelihood um so now you know i
found a set that actually is still at
16. so the thing is still
going for a bit to converge and down
here
these are the posterior deviations and
again ignore the
ignore a lot of the units here but the
way that you can read this
is just that 0 here corresponds to the
prior values
that you set um and if it's going
if the bar here is going down then that
means that the

posterior parameter estimate is uh
is lower at this point it's gone down
from the prior value
and the this kind of red pink thing
around it that's the posterior variance
um so in this case and parameter one
here
is uh the action precision and parameter
two is the
reward seeker reinforce or risk seeking
i think i can double check whether those
are backwards or not um
but uh which order those are but this is
just saying that whatever this first
parameter is
um the the posterior mean estimate here
is
um you know whatever that is that
actually corresponds to the real units
of the um
of the uh parameter but it's not very
confident
in that posterior immune estimate
whereas here
the second parameter um it's also this
negative
posterior at the moment but it's very
very confident in that posterior
estimate
um um so so that's what that means and
these will update
with each iteration um but so now you
can see that actually the thing kept
exploring
and now it's actually found a set of
parameter values
that um you know continues to actually
explain the behavior

the simulated behavior quite a bit
better right so now it's at like
negative 13.

so you're gonna see this it almost kind
of converged for a bit but now it's
actually kind of going back up
um so eventually eventually it will
converge but it's just doing really
really well

at finding values that explain behavior
uh well um one thing i should point out
though is that the actual values of
these absolute value of these are not
really informative because they're
they're they're basically the sums of
the action probabilities

for each trial so these numbers will be
bigger if the task has more trials
um so in and of themselves it's only the
relative values that are meaningful
um this reminds me actually it reminds
me a lot of

bayesian methods in phylogenetics
where it's like what is the likelihood
of this phylogenetic tree

it it's given some number by a program
which sounds weird to think about the
state space of all the possible trees or
something like that

but it turns out by doing this kind of a
gradient descent

searching through all the possible trees
you actually do

converge on a tree that is
generative of the kinds of data that are
observed and so it's just really
interesting to see how this is working
and it's fun to watch the number

drop down to so
so now so in this case it converged for
the first person
um and you know these were the posterior
deviations and then here's the simulated
behavior
of the participant under those
parameters
so you can see that you know the
parameters do
fit the behavior really well right the
probabilities are pretty high
under those parameters for each for
every action
you know there are some little
exceptions but but it does pretty well
um right so now it's just now it's just
moving on to the next person
sorry what oh it might just be helpful
to describe what the blue dots are
just for yeah sorry again i'm assuming
people have
watched video sessions so blue
corresponds to the actual chosen action
um on the first at the first action for
each
participant um so basically this is
saying and so black
means probability one and as it goes
toward white that means probability zero
so this is just saying basically 100
probability under the model
that the agent would choose the hint and
the blue dot says that's what they
actually chose
and so on and so forth so anytime that
there's a
light gray you know that's on the part

where the blue dot is that means the model didn't really do that well at predicting that behavior um but you can see in this case most it's pretty dark around most of these blue dots so it does pretty good um is the point um so now like i said this is just going to iterate around for the six agents that i mentioned and it's just going to compare and then it's going to compare the three uh the three that did have a learning rate and the three that didn't have a learning rate in the model is what it will do um and so just to kind of summarize what you've said so far so the steps would be something like you would actually get your empirical task right and think about how the sub the actions that the subject has actually got or the participants actually going to make relate to what the model is going to make all the actions available the model you come up with some mapping and then you would then presumably in this just translate them in some way and plug those into your u mdp dot u which is the actions participant shows and mdp.o which is the observations that they actually saw yeah and then you would plug into the algorithm essentially yep you plug it into the algorithm and um

and then it just the algorithm just repeatedly computes the sum of the log likelihoods for each trial um and then and then tries to find a minimum of that sum okay and so in terms of this is a something i've wondered about um how do you choose what the best trials uh or what the best priors are is it okay just to kind of in in silico uh simulate a bunch of things and just say hey this this seems reasonable um in a lot of cases yes i mean so so there's a couple things to do one is you can you can do the kind of modelist or the recoverability stuff um you know beforehand and try to find a set of parameter values a set of prior values for which the um the the parameter estimates are recoverable right so if you set one set of priors then maybe the thing does get stuck in local minima but if you said another set of priors then the generative parameters um actually do end up matching the estimated parameters really well um so so there's a couple different things so one is yeah do some of the simulation recoverability stuff ahead of time to find good priors another thing is you know if you just you know estimate participant behavior

and you start to see
that the posterior estimates tend to be
really far away from the priors you're
setting
um that's kind of an indication that
you're probably not setting very good
priors so then you might like try
setting new priors that look closer to
where everybody's kind of moving um yeah
so we do have we do have a footnote
about this in the paper from memory
um but it'll be good to kind of make
that explicit
another way that i've seen that in the
field of evolutionary biology is
if you have wildly disparate priors like
the so a fallacy is that the uniform or
the flat prior
is like unbiased there's so much to say
about that i'm sure
ryan knows well but like if you have one
prior that's stacked up against zero
and another prior that's stacked up
against one and then they both converge
like from kind of multiple sides
that's a simple example but if different
families of priors
and models that are very um paired down
ranging to ones that have very complex
error
models if a lot of different layers of
complexity of the model
and densities that start stacked up
versus one end versus another
then of course that's a difficult thing
to manage but that at least means that
given the empirical data you have and
the task that you're modeling

you have a really predictive model so
again we're not realists we're not
actually getting the truth with these
recursive and iterative and
multi-perspectival models
we're just fitting more and more of the
variants explained
in our empirical data yeah
yeah so just uh just to kind of give you
guys another example so you can see
now for this other set of parameter
estimates you know convergence took way
less time
right it just took you know seven
iterations and it converged really
quickly
and one reason to see why is that for
this one it didn't have to move
values very far from priors because this
was an agent
whose behavior was generated by
parameters very close to priors
what one other point there though is
even if it looks flat for ten time steps
it could still be trapped which is why
things have to be seeded and have really
good randomness
because there's no hard and fast rule
for when you terminate
so for example in a lot of the
evolutionary simulations we would do
you discard the first like 10 000 or
more
time steps like burn in you just discard
them because you think that actually
it's too much reflecting your prior
estimates
and then even if it looks like it's

converging or as one of my professors
called it a fuzzy caterpillar
because it's kind of like the model was
sampling from the fullest possible range
of variables
and it was still coming back home that's
the fuzzy caterpillar
but even then it could look like a fuzzy
caterpillar for like a lot of time steps
and then just totally
hit on a new combination break into a
new realm so it's really
hard in a challenge so just so people
know that's actually not
true in this case so variational base is
a is a deterministic in the sense that
um you you don't do this kind of you
know
burn in you don't um right variability
you just deterministically like if i ran
this over and over again it would give
me the exact same parameter estimates
each time
um so there is there is actually no
variability i mean the kind of thing
you're talking about is more like monte
carlo
yeah it's gonna say sorts of approaches
yeah sorry i didn't mean to say that
there was a burn-in
um i agree that was just yeah little
analogy but thanks for clarifying so one
advantage of monte carlo like sampling
methods is that you over
given infinite time you're guaranteed to
margin to approach
the true posterior or to obtain the true
posterior

uh it's also extremely computationally expensive

um variational base is really quick like you can run this stuff on your laptop so um the monte carlo is based upon sampling that was the different domain that i was um referencing there you have to sample

in cases where you don't know some of these distributions don't have them defined so perfectly

but when you have access to this level of specifying

then there's a whole new range of techniques which is why it looks more like matrix multiplication

and this sort of like convergence to the variational

free energy estimate rather than just sampling endlessly from a landscape that's of unknown

you know anything else but yeah thanks for that sorry

so so just take i mean show again just to give you guys a sense so in this so this kind of participant you can see that it's pretty kind of

the distributions are a lot less precise and this is a person who has a much lower action precision value

so these were generated by a lower action precision value

agent and the model is doing a pretty good job just finding a low action precision value that kind of spreads the distributions

flat enough around it that it captures it provides decent evidence for each

action

um and so anyway so just just to uh
again to to give you guys an intuition
for what's going on

um but so once this is done

then what will happen is we'll have
these

uh the free energies of the winning
model for each person for each of the
two models

um the one with and the one without
learning rate um and so we want to do is
we want to do bayesian model comparison

um which is where you're going to
compare the free energies for each model
for each participant

to find the model with the lowest free
energy across participants

and the winning model um so the the
little spm function that you know we
include here

it'll spit out several things but the
one that you want to focus on

um just to keep things simple is um
the one that has the highest what's
called the protected exceedance
probability

um and this is just the probability that
each model

is the most likely model across all
subjects while taking into account the
null possibility

the differences in model evidence or due
to chance so

so it's like i said it's just it's just
which is the best model

um when taking into account the null
model

um as a possibility um
and i'll um i'll show you that
um in a second let's see is this thing
it's still going
it's uh yeah okay still going
um hopefully it'll be done soon but so
i'll keep i'll keep moving through here
and then we can return to this once it's
done
it's actually going pretty fast since i
made it so few trials
um but so so that's you know what we're
doing here
um and uh so now
you know we've kind of already touched
on this a bit but
so now is the point where we would
confirm
that the best model is identifiable you
know or that the parameters are
recoverable
and this is so i already mentioned this
a little bit but that you know it's it's
clear that not all models are
necessarily going to have
unique parameter solutions right so if
you have a landscape like this
you might start out priors at very
similar places
and gradient descent would lead you to
minima
different minima you know different
combinations of parameter values that
are equally good
at reproducing a participant's behavior
so so you always have to show
that whatever model you're using and the
priors that you're using

will give you the same parameters
that you fed in to generate the data to
begin with
and so here um again we've already
talked about this a little bit but step
one is
you would specify multiple sets of
generative parameter values which is
what we're doing
um and this is important as you should
select values
that are the same or similar to the
actual parameter combinations
that you got in your in your true
participants
um because it can be the case that
parameter estimates are totally
recoverable
for certain parameter combinations but
are not recoverable for other parameter
combinations so you care about the ones
you're actually getting
for your participants um
so then you want to simulate behavior
so generate simulated behavior using
each of those parameter value
combinations
run that simulated behavior through the
estimation routine
just like you would for real data and
then check whether the
generative parameters and the estimated
parameters are highly correlated
so when i run this in advance
you know so without having to run it in
real time like i'm doing right now
then for this particular case and bear
in mind this is when i'm using 32 trials

not like the
16 or whatever i put in now um just to
make things go faster
this is what i get for it for alpha for
action precision for the two parameter
model
so you can see even with only six people
the correlation between the
generative uh action precision and the
estimated extra precision is pretty good
it's 0.81 so even with six people
right it's significant um
whereas for uh
the only one well anyway i have a i have
a bunch of them i can i can
i can pull uh i could probably pull them
up um
let's see that's for a totally different
thing
um cool
that was very tutorial
interesting uh yes sorry i'm just trying
to find um trying to find the uh
the figures that i have it's probably
here um
well no okay not positive where i put
them actually
but if i can't find them here then
uh it will it will spit them out
um but um but
point point being um these are uh
um in this case even with just six
values um
all of the parameters um tend to be
really good
right in terms of recoverability the you
know the correlations between generative
and estimated parameters tend to be

you know between point seven and one
point seven and point
nine something or other um so they're
they're
they're good um so then so then finally
the last thing that we wanna do is once
we have
parameter estimates for each person in a
winning model
that is recoverable then we can take
those values
and we can put those parameter we can
use those parameters
as individual difference measures
between subjects um
you know at that point there's a number
of things that you can do
so you know the kind of simplest thing
you know if you if you want to kind of
fall back on on frequentist
statistics is you know you can do
standard you know t
tests anovas correlations regressions
etc um
and um and you know i mean you know
we've done that before
it's fine in some cases um if you want
to kind of stick
with the more kind of general bayesian
theme of active inference then you can
also do things like
um you know do things more like base
factor analyses
and you know both of these are often
kind of used
or or even used together um
and just to give you a couple examples
sorry i'm just going to triple check

that this thing is not
done yes still not done just one note
there is the spm textbook
has many points of contact between
parametric
and um like standard very classical
statistical approaches and then very
bayesian
approaches and mixed methods and it will
switch out or show it both ways
so it really is pretty interesting how
it comes together here
so so to kind of show you guys a couple
examples of how you would use
um these kind of you know use parameters
at the group level
um you know this is just i'll give you a
couple examples from from our papers
um so in this in this one in journal of
uh psychiatry and neuroscience
um what we did is we had a this simple
kind of approach avoidance conflict task
where more or less the participant had
to kind of choose to move this little
avatar guy
closer or farther from one of these two
ends
of this little kind of runway thing
and they knew that the closer they were
to one side the higher the probability
was that they would get an outcome
associated with symbols on the left side
and same goes the closer they are to the
right the higher probability of getting
the outcomes associated with the right
side
i mean here a rain cloud means they
heard like a really aversive sound and

saw like a really aversive picture
you know like it's like hearing girls
screaming and seeing a picture of her
being like pulled into a van
you know like really negative stuff um
and uh whereas the sun thing here man
you kind of saw this kind of like
neutral maybe slightly happy thing
um and uh you have this kind of red this
kind of bar thing on the side and the
more
filled the red bar was the more points
they would win
um so you have kind of clear cases right
where it's just if it's just rain cloud
versus sun
you should just go to the right if it's
just sun
and sun plus some reward you should want
to go to the right
and then you have these conflict cases
where it's negative stuff
plus you get some reward negative stuff
but it gives you plus you get even more
reward and negative stuff and even more
reward than that
um and so and so that's how the task is
structured
and so you can make a pretty simple
model of this
where one state factor is beliefs about
the trial type
right whether it's uh this kind of trial
this kind of trial this kind of trial is
controller that kind of trial
um and then you can have a state factor
corresponding to
beliefs about the runway position right

so uh beliefs about whether the avatar
is in position one two three four five
etcetera
that's it and then you just have um
you set up likelihoods that generate the
probability
of um of each position
generating what sorts of outcomes and
what sorts of runaway positions
given beliefs about trial type right
here ryan so you have it phrased as
there being five trial types but
would it be possible to have like a
model where it was like a state
estimate left and a state estimate right
for
the stimuli type and a state estimate
left and say estimate right for the
reward value
because this frames it in a very
behavioral trial
centric way with doing estimate on which
one of the five scenarios you're in but
i'm just trying to think about cases
where
you might not know which scenario you're
in or even what the total set of
scenarios is
so you're just doing state estimate on
reward and on stimuli
um yeah i mean i mean yes i mean the
well
well i guess it kind of depends right so
i mean if you if you were to
give somebody sort of uncertain cues
about trial type
um then um then
i mean presumably you would just you

wouldn't need to i mean you would just still have

um if they know what the different trial type possibilities are then you could just have uncertain cues so they just don't have a precise belief a set precise prior belief over the different

over this state factor but if you didn't if there wasn't any set beliefs about trial types and there's just any possible combination

of you know sun clouds zero points two four and six

then um um i suppose you could do something where you just have a kind of non-factorized state factor that has like

just every possible combination or something like that um you could do that i mean it would uh

um there are reasons to actually build like so there are

obviously time saving and model building considerations that mean you should use factorized distributions

um but there are actual empirical considerations which means that i think if you want to build a model that's kind of realistically how the brain works

you should tend towards in a lot of cases you should tend towards factorized distributions

so for example there's good evidence that there's a factorized representation of task phase

in pfc and mtl uh there's pretty good

evidence so

i mean the what versus where streams

in vision that's a factorized

representation

um so very

yeah so you're kind of using it in like

a coarse graining

sense to say sometimes you don't want to

even

allow for the all by all because it's

more categorical

how the task is being modeled anywhere

like fight or flight

you wouldn't want to have all

combinations of elbow and knee movements

you're fitting something that's kind of

at the wrong dimensionality

and so for a lot of reasons not the

least of which it seems like that's what

organisms would do would be at that high

dimensional manifold or the factorized

representation

yeah well and and in this case right i

mean just we have experimental

the experimental design is such that

they are they do know ahead of time what

the different possible combinations are

right so i mean they just they just

already know that there are these

combinations

um so it's it's consistent with their

beliefs about the generative model that

we have given them

via the instructions um so

um so so in this case

um you know we had a a pretty uh typical

you know sort of sort of model graphical

model with two parameters

we had our the beta here which
corresponds to the
expected for energy precision um here
you know just to be intuitive for the uh
clinical audience we were going for we
just called this decision uncertainty
um and then we had a and so that's just
this is again
this is the rate prior um or rate
parameter for a gamma distribution
uh over uh over this uh
gamma term um and uh
which again is a uh a thing that
modulates the
expected free energy estimate over
policies
um so it modulates this g thing um but
then the other parameter we had
was this uh emotion conflict parameter
which just said basically how much
they uh dislike the negative stimulus
how
aversive they expect the negative
stimulus to be um
and so you can kind of show in
simulation so each one of these uh
vertical bars here corresponds to
beliefs about
uh the what state you were in on the
runway um
so if beta equals one and uh
the um and uh the emotion conflict
equals one
then you should expect that if the you
know good thing is on the uh
oh it is okay and this is a this is a um
conflict plus
two points uh trial um so basically the

thing will approach the reward even if it's going to see something negative if vc is one and it will do it deterministically whereas if ec is three then it will fairly deterministically choose to go away from the negative stimulus even if they would get reward whereas if they have higher values so more decision uncertainty than this distribution becomes they become a lot less confident in this distribution and end up choosing these kinds of values that are more like in the middle um and um and uh the likelihood is is you know pretty clear it's just in each trial type each of the different um each column here is a different runway position they will just generate the negative stimulus with an increasing probability as it goes left and the positive stimulus with an increasing probability as it goes right um etc uh the only confusing thing here is that white means a higher probability in these whereas black it means higher probability here but um but anyway so that's that's it and just kind of etc etc for the five trial types so in this case what we found was if you look at healthy controls versus people with depression and anxiety versus people

with substance use disorders
the emotion conflict is actually
interestingly higher
in healthy and lowest in substance use
disorder
whereas decision uncertainty is highest
in substance use disorder
and kind of medium and depression
anxiety and low
and healthy controls and these are
significantly different
and that's true in a propensity match
propensity match sample and in a
larger sample so this interesting thing
where
it might be more of a what might be more
clinically relevant as this kind of
uncertainty over options as opposed to
just
being more sensitive to negative stimuli
um right how would you say how would
that shape just kind of curious if
it doesn't have a if it's not known but
how would that shape
treatment or conversation or approach
in a given situation from whichever role
makes sense like to say oh it's not
actually
this psychological construct but it's
actually
related to something like this
um i mean in terms of like informing
treatment or something like that
sure like treatment via any modality
um yeah i mean so i mean typically the
the main source of things you might want
to do is you know either talking about
um

so say you know at baseline before
somebody starts treatment
what their beta value was right
it might be the case you'd have to study
to show this but it might be the case
that
given different beta values at baseline
you know
like people with high beta values might
respond well to cbt
cognitive behavioral therapy whereas
people with low beta values might
respond better to act or might respond
better to
an ssri or you know like something like
that
so either it's either it's something i
mean that's kind of the ideal thing is
you want to say
can i get this information and it will
give me information about how to treat a
person
um but there are lots of other things
you might do besides that but that's
that's like a primary kind of
you know like ultimate goal example yeah
it's interesting how there's probably a
lot to be
said and learned about the actual
application of active inference
clinically
but even here on the beginning of
applying it
we can use it as just a potential
biomarker
just like a summary statistic related to
a questionnaire
or some other thing that's estimated

from empirical clinical data
fitting a different kind of underlying
generative model so instead of doing a
regression on people who have
higher on this end up doing better or
worse in this kind of a program
well now we can just do that same kind
of parameter
testing in the context of a different
type of model it's an active inference
generative model so i think that those
are just some points of contact but it's
really interesting stuff
yeah definitely i mean you can do lots
of other interesting things with these
right so this
this beta parameter if you remember from
previous sessions is what's associated
with
um it's proposed to be associated with
um
dopamine dynamics in uh
in the brain and i'll show you briefly a
a study
um an example study later which
where philip shorten back like i
mentioned before actually showed that
the
trial by trial updates in beta that are
predicted by the model
um were correlated with a bold signal
in fmri in
in the midbrain in a midbrain region
that's associated with dopamine
you know so whereas whereas the way that
we're doing this here we just have a
stable beta estimate for each person
but you might look at um say individual

differences between
contrast values in an fmri analysis at
the group level
um you know so like people with higher
beta values have higher
say uh basal ganglia
uh responses to reward versus no reward
or something like that
um so you could do that kind of thing as
well so both both individual level and
group level um
fmri sorts of approaches as well as
much fancier things but those are just
two kind of simple examples

[Music]

so as another example of doing this in
the in the domain of
kind of perception instead of decision
making um
you know we we used uh this is in the
context of a task
where a person is just told to uh push a
button every time they feel their
heartbeat
um and um so in this model uh
we don't even have like an explicit
policy selection part
um all we do is they start with a
precise belief that they're in this
start state at time one
and then they have prior beliefs about
whether or not
they're going to transition into a state
of feeling their heartbeat versus not so
probability of no heartbeat probably of
heartbeat
and those priors are encoded in this in
the

b matrix here the transition matrix
where sort of the higher the ϕ is the
more they expect to feel a heartbeat you
know more often
right on each trial um
and um and then also there's a precision
value here which corresponds to
beliefs about how precise the actual
afferent signal is coming up
from the heart um and so we can estimate
this ip parameters intercept of
precision and this
prior over heartbeats um i should say we
also compared this to a model that
included learning
in this task and in this case the model
including learning didn't win
um and so what we found here is
and i should say also that they do this
task three times they do it once when
they're told they're allowed to guess
once when they're told they're not
allowed to guess and only to press it
when they're sure they felt something
and one where aside from in addition to
no guessing they're told to hold their
breath which kind of makes it
um on average easier uh to feel your
heartbeat
um and um so it kind of like amplifies
the afferent signal makes it more
precise
what we found was that in healthy people
the breath hold actually amplified into
receptive precision a lot
whereas in all the clinical groups
anxiety depression comorbid depression
anxiety eating disorders and substance

use disorders they just stayed flat
low intercepted precision values
so that the actual changing the
actual precision of the afferent signal
didn't have any effect on their beliefs
about the precision of the signal
um you know so it's something like like
a rigidity in the way that the brain
treats
uh afferent interceptive signals and
psychiatric disorders trans
diagnostically
uh and so again this is just a again
kind of a standard like mixed model sort
of analysis
um whereas if you compare say
estimates for prior expectations
everybody
um showed higher prior values in the
guessing condition
than in the other two conditions which
you would expect
but there were no differences between
healthy and clinical groups so it's kind
of interesting it says hey maybe this is
more a precision issue than it is a
prior expectation issue
in terms of clinical significance or
another task or another task has to be
explored
chris put in a plug how cool that paper
was i think there's
been a lot of conceptual work on
interception active inference and
there's been a lot of discussion about
whether it's priors or precision or
anything like that
um and to my knowledge i'm happy to be

corrected

about this by ryan but um that was the first time that

it's really been tested empirically

right uh

yes there's no there's no other there's no other

like actual formal fitting models to data

um studies that have tried that before nice epic work

there are there are papers that have tested like more kind of like

quote-unquote qualitative or just kind

of like go up versus down sorts of

predictions that fall out of

computational models but

not actually fitting a model yes this is

the first model based analysis of this

kind of thing right

yeah and we've um and it's cool we've

actually replicated the results in

healthy controls in a second

uh sample now so it seems like the

effect at least the effect in healthies

is pretty robust

we haven't been able to replicate we

haven't tried to replicate the uh

the lack of uh of an effect

um in uh clinical populations yet but

that's kind of in the works

um cool so

so so anyway so this continues this

continues to go here

yeah but it should be i think it should

be almost done

pretty quick it only does six people i'm

surprised

an interception question and then one question from the chat so what other interceptive methodologies might exist so you did a heart rate estimation task what other interoceptive modalities might be amenable to this kind of quantitative analysis um so definitely um definitely cardiac interception tasks are um are uh most common just because it's it's actually quite difficult the methodologies is it's pretty difficult to um you know use so for instance in vision right you can like very tightly control or in audition you can really tightly control the timing and magnitude you know of like the of the the input signal right whereas it's it's hard to say precisely control um changes in the signals that you're getting from the inside of your body um so so it's um you know it's difficult at least the heart is a kind of like signal that has a rate and um and you kind of know precisely when it sent the signal upward you know and things like that so they're definitely the most common there's lots of different ways that people do cardiac there's lots of different cardiac perception tasks a lot of them have come

under

a lot of recent criticism recently um

for various reasons

um there's been some studies that show

that

um for instance like standard uh what

are called heartbeat counting tasks

where people are just kind of asked to

like over a period of a couple of

minutes

just count every time they feel the

heartbeat and you just kind of

look at how close their counts are to

the true number of heartbeats

um there's a there's a number of papers

including our paper actually that show

that this looks like it's primarily just

tracking

prior expectations um it doesn't it

doesn't tell you a lot about

you know what the the way that they're

actually treating individual signals

um and i mean in ours in ours we

actually showed this that um

that if you used a standard heartbeat

counting measure instead of the

heartbeat tapping

you know which was our primary measure

that we fed into the model um prior

expectations in the model

predicted the heartbeat counting uh

like accuracy values like 0.9 something

so so it was it was like a nice

confirmation that yeah this is

primarily about price

it points towards having a generative

model that can then be deployed and

modified and corroborated across

settings because then you could say well what would be the task or what would be the trial and set structure how many participants would we need to resolve a parametric difference of such and such so statistical power down to a lot of other features would be influenced okay let me ask this question from the chat so that we do hold on i need to finish that previous question um so so but aside from cardiac stuff there are a number of other methods um there's so one that i currently do in my lab is something called i'm using like breathing uh breathing resistances so basically you can you have these people wear this little kind of darth vader mask kind of thing and you can change in really precise subtle ways how is how much resistance there is when they try to breathe in um and you can um and you can get sort of individual differences in sensitivity you know where some people you know feel uh that change in like the how hard their lungs are needing to work basically um at different at different loads of different resistances than than others um and you can also use it as kind of like an anxiety induction which is how we use it to try to you know precisely induce uncomfortable intercepted states at different uh intensities and see how that affects

things what about muscle there's also
sorry what what about muscular like how
heavy is this or how what angle is your
arm at
um i mean we don't consider that
interception that's more like
proprioception or
um or some other sensation um but
certainly i mean there's lots of tasks
like that out there
um the kind of thing that that you know
we mean by interception is things like
you know feeling what's going on in your
stomach feeling what's going on in your
you know your heart or your lungs or um
you know like various like
effects of hormone levels you know like
stuff going on
inside cool um so um i mean i should say
i can't talk too much about it
but we have actually developed a method
for
um precisely inducing inducing with
precise timing
um sensations in your stomach um and
we're currently just finishing up a
paper on this uh using that method with
eeg
um to test some predictions of the
neural process theory for uh
uh gastrointestinal interoception so
that's another thing that's kind of
common looking forward
let me ask the second question if it's
okay yes
what are the similarities and
differences between the active inference
model

of emotional inference was how it was
phrased

and lisa feldman barrett's approach to
emotion construction so if you don't
know

once i i i just wanted i'm just reading
it off but um

[Music]

that's probably that's probably a
question for another time i mean for
people who are familiar with
you know my actual like simulation work
in this area i've

you know i and my group have published
multiple

active inference simulation papers
specifically about emotion

inference um and um

those uh models to the so i should say
it's not a straightforward question

because you could build

active inference models that are uh
that would do more or less something
constructionist

or you could build an active inference
model that does something

less constructionist so so it's it's
actually not as though

um active inference has something really
unique to say about

whether or not constructionist views of
emotion are right um

but um but the um but i should say that
the the most straightforward models

um that have to do with uh

constructionist views where where what
you're trying to do is

learn and infer emotion concepts um

those the most straightforward ways of
doing those are via the sort of
multimodal
inference inference process that
definitely has a kind of when those
mappings between
emotional concept like emotion state
concepts and um
lower level you know sorts of like uh
observations like for instance like
observing your arousal level you know
observing how negative or positive you
feel things like that treating those
lower level things as observations
um when the likelihood between those
and um emotion concept states at a
higher level when the mapping is in the
likelihood is uh
probabilistic then you end up having to
learn probabilistic mappings that look a
lot like
constructionist types of inference
stories
um but what's kind of less clear is is
that um
constructionist views say don't say very
much about
how the kind of states in your body are
generated in the first place
um you know say like you know if i see a
predator or something and i have
i have this big change in my heart rate
and like muscle tension and things like
that
um and then i feel that and i infer
construct
right some kind of belief about what
emotion that corresponds to in this

context

there's a you also need a story and an active inference model for the mechanisms that generate those responses before you make sense of them and you know we've we've talked about doing things like that with active inference models um and it's not so clear whether or not something like that would be consistent or inconsistent with um

constructionist views um but uh but yeah anyway

um so hopefully that helps the active inference doesn't solve the does it doesn't uh take sides necessarily i guess that's my point really interesting i didn't know about that um

theory so it sounds like something where we can use active inference to construct various kinds of models so all these orthogonal models and we're like looking always at the two by two on the active stream is it going to work on this axis and this axis which debates are relevant which ones aren't for active inference and the free energy principle so you want to just carry on a little bit yeah i mean so

i mean just to just to say i mean because this this actually comes back very well to the to the actual topic of you know today's session um is that to really solve these questions what you need to do right is you need to construct an active inference model

that looks very constructionist and an active inference model that doesn't look very constructionist and see which one fits empirical data better right so um see it's more kind of a way of precisely testing uh different you know hypotheses of different models as opposed to you know saying this model must be correct theoretically yeah i think it's kind of important to when you're thinking about this kind of stuff people often talk about like active inference as a theory and as we've said like multiple times there are multiple process theories that could fall out of this i think it's a very general framework under which you can build multiple hopefully competing theories um so yeah i mean and in addition with computational psychiatry stuff you could just use you could think active inference is literally false like as a theory but it would still be you could still also simultaneously think it's an awesome individual differences all like that would be an odd set of beliefs to hold but there's no contradiction there yeah um yeah no for sure uh yeah i mean this comes back to the you know like uh whether you think of models this kind of model is getting to like the ground truth versus just using it you know instrumentally um but uh but

yes

um so okay so so last kind of thing here
is is that

um you know in addition to being able to
do standard you know sorts of like you
know again like anovas regressions you
know frequencies approaches or even
you know base factor analyses um
in place of or in addition to them um
one thing that's really nice about
getting parameter estimates the way
we've been talking about is you don't
just get um point estimates you don't
just get posterior means

which is you know what we've been
talking about you know with like
analyses like like that

right like this is just you know means
of posterior means

um whereas um

the active inference to the parameter
estimates in variational bays

also uh correspond to posterior also
have posterior variances

right so it's not just the mean but it's
also the confidence that the estimation
had around that mean um and so that's
actually additional information that's
useful

to incorporate when you're doing group
level analyses

and and

parametric empirical bays which is um
something that again

was initially developed and is primarily
used for dynamic causal modeling

um happens to work really well

um with the uh with the the setup for

getting parameter estimates
and for the way that we get parameter
estimates again like i said before
already uses dcm scripts um so you can
actually do

these parametric empirical bayes
approaches that are more or less
general linear models but that also use
the posterior variances
um to get kind of um probably the most
principled

you know way of doing you know like the
fully bayesian

um group level stuff that you know kind
of thing that you could do
with these individual level parameter
estimates incorporating both the means
and the variances

um and so uh i think this
is this thing done

okay this has got to be the last one
but uh but um anyway

uh i had there is a thing that i can i
can if it takes too long i do have saved
versions of this i just wanted to show
you

uh um i was hoping to be able to show
you the actual model comparison part
um but so you know as an example of
using parametric empirical bays
in this paper on substance use we did do
this

so in this model in this task it's
pretty simple

all people do is they just um repeatedly
choose option one two or three um
and on each one either a green ball
falls or a red ball falls

green all means green ball means a win
and in advance they don't know
which of the options has the highest
payout probability
so again it's explore exploit right so
you have to kind of try out different
ones until they become confident which
one's best and they kind of keep picking
it
and in this case we were able to
estimate a number of parameters
we estimated action precision um we
estimated
that risk seeking parameter that i
mentioned
um we estimated uh separate learning
rates
for um when they had a win versus when
they had a loss
um and we're also able to estimate this
kind of a information sensitivity
parameter
um it's probably too much to explain at
the moment but it's kind of like a
belief rigidity it's like an initial
how uh the higher it is the less you
should think you need to seek
information
um basically um
and what we were able to do with that is
um over here we did standard
kind of group differences and you could
see that action precision was a lot
lower in substance users
and um learning rates for uh
losses were uh lower um but we also did
the pebb version the parametric
empirical base version over here where

we could get these group level
uh effects that included the means and
the variances
and um and you could also see that these
effects

were were there and you could actually
show like the posterior probabilities
um and things like that um that were uh
again it's just a it's just a kind of
nicer thing to do that would give you
um that'll give you additional be able
you'll be able to incorporate additional
information and keep things fully within
a bayesian framework

um and um and so
if this thing is still are you done
uh i have another question we can ask if
you wanted to just let it run
on this all right so just curious
christopher said that he'll be using
this in his phd

so what is your phd project or i know
it's
starting the phd and everything but what
are you thinking about using it um
that's the first question and then
there'll be a follow-up part

um so i work in alex wolgar's lab
who's a pi at the mrc cognition brain
sciences unit and her lab studies
cognitive control

so i'm planning on doing a lot of work
on cognitive control
um at the moment the
so i don't really want to say too much
about what projects i have
yeah most because just what questions
are you curious about or what

do you think are exciting so mostly i'm
really curious about using
or using active inference to help us
think about
and formalize some otherwise kind of
less formal hypotheses in the realm of
cognitive control
and then using those kind of generate
some testable predictions
um cool although yeah one thing i should
say is like
i don't think the active inference is
always the best tool to answer all
questions you might have
depends on what level of analysis you're
working at i happen to be really
interested in like algorithmic level
questions where active inference is
really good
um if you're interested in
implementation level stuff this might
not be the best tool for you
great and so the second part of the
question was are there any
fep or active oriented supervisors at
cambridge but
if there's yes or no that one but more
generally i'd like to hear both of your
perspectives on like
let's say somebody were starting
research as a graduate student
in a lab that wasn't in this actual
line of research already so what would
you convey to a starting grad
student or starting researcher who was
looking for a phd mentor
or who wanted to do a research program
that was like aligned with a lab that

was studying something cool
and interesting but they wanted to take
an active inference perspective on it
um okay just so just so people know it
is now done yeah
yeah but but go ahead if you want to
answer that question quickly
um i was just gonna say i have a couple
things to say i think it just depends on
your personality really
uh are you the type of person who
doesn't mind doing a lot of things
alone and without much help in which
case you can do something
that you or other members of your group
aren't doing if that doesn't bother you
then go ahead
this tutorial is a great resource um
if you are working in a lab i i would
genuinely just try and find a lab that
studies what you're interested in
that's probably the good advice i mean
so one thing to say is like ryan is
i haven't signed the paperwork yet but
um he's going to be my associate
supervisor
so i do have someone in my like
supervision group who can actually help
me with this stuff and so i would always
say make sure whatever you're doing you
actually have people who can help you
great point and i wonder if in the
remote world
it will be easier to say i i remember it
was kind of a big thing to have
i didn't even have this but like
somebody who's a committee member from
another

university so now it might be a little
bit more possible to connect remotely
anyways
ryan i'd be curious to hear your
thoughts on that and then we can go back
to the model
um sorry i don't think i fully uh just
suggestion
a a starting researcher somebody who
asked to be in your lab but they were
you know there's not room for them in
your lab or they're in a different lab
already
but they want to do something with
research
in this area oh i say i see yeah i mean
so that's a very tricky question
i think i mean when you're deciding
where to go to grad school
i think it's really really important
that
you find somebody and that you're
confident you found somebody
um where you uh was a good match between
you and your
and your supervisor um you know i've
seen cases where people
start in labs and it's just a not a good
match between your supervisor's style
and your own style as a student um and
you know a lot of times that's a recipe
for either a really tough time getting
through grad school or even deciding
that maybe you don't want to be in
science anymore um i've seen
i've seen that personally you know like
the case with other people whereas if
you find somebody who

you know kind of matches your style
right in terms of like
how much direct micro management
supervision you want versus how much
independence you want
you know like having a supervisor that
really allows you to be creative and
come up with your own projects versus
someone who
you know would prefer that you're
working a lot a lot of stuff that's sort
of directly
um sort of more narrowly in line with
what the lab's already doing
um you know i mean all sorts of
supervisors
will have will have
you know types of grad students that
work well with them but
finding the right matches is really
really important
and part of that you know does also
correspond to common interests
um i mean if the question is you know i
absolutely cannot find somebody
who does what i want to do that is
really tough um
um but i would say in that case uh you
know what you were mentioning about um
you know kind of uh remote or uh
co-supervisor sorts of situations where
as long as you can get
your primary supervisor who's a decent
match to you in general
to be okay with you having some other
supervisor
um or mentor of some sort that will do
the sort of thing

that you want to be doing
methodologically and there's enough kind
of overlapping interest between
real you know primary mentor and
co-mentor then i think that can work
well

um you know in those cases but um but
if at all possible identifying a primary
supervisor who does that
the methods and is associated and has
the same interests um
is you know definitely the best option
if you can

um great advice but um
okay so all right so this is now done so
you can actually see what the script
does once it's done um
so you can see so what it will spit out
um

i i should say this is also a new
feature i just added this
to the uh to the to the tutorial script
to this main tutorial script so
if you go to the github and download the
newest version then it will generate
these

these uh scatter plots for you um but
um so it will show you for instance so
when we only did 16 trials right
then recoverability for learning rate
for the three parameter model wasn't
awesome right the correlation between
true and uh

estimated parameters was only 0.58
whereas for risk seeking the correlation
was 0.95 so super recoverable
and for alpha it was um

i don't know it's not showing up but it

was 0.42 so not
great um whereas for the two parameter
model
risk seeking was 0.95 and alpha
was 0.94 so i should say
depending on the values the r and p
values might not show up on those graphs
so i have it set up but it will print
them out for you here
so alpha recoverability 0.94 p equals
.046 etcetera
so it will give you all of that it'll
just spit that out for you
it will also tell you the average log
likelihood
under the two parameter model and the
average
action probabilities um of the two
parameter model and the same thing for
the three
so the fits are actually very similar
in terms of the log likelihoods and the
um
action probabilities and so
when you actually calculate the
protected exceedance probability
which it also will do for you p_{xp} um
it will show that the um in this case
the um
the second model so the model with
learning has a
uh has a higher exceedance probability
than the first model
but this is actually not necessarily a
very clear winner here i mean what
you're hoping for is
something like one and zero or you know
point point eight point two point nine

point one

something like that um so um

and again this is just an example this

is not going to be reliable at all

because i did

16 trials right you know if you did a

real task if you just use the 32 trials

that i actually have in the

in the real code on the github um

it will be better than this but you know

most

actual behavioral tasks you know like

for computational modeling studies are

going to have

you know like 100 you know or more

trials um to really get kind of precise

estimates and and know how good these

models are actually doing

um so but um so

so that's the kind of thing that you get

here at the end so you'll get

so i should say this trajectory thing is

not that meaningful it's just showing

for the first and second parameter what

the trajectory of the change in the

parameters are

as it converges toward the posterior

values but when you have more than

two parameters it's not that informative

um so

so um you know we have in the paper

you know we have this figure that more

or less just kind of depicts

an example of what this kind of thing

looks like um

and more or less as long as this kind of

you know iterations converge and has

this kind of nice increasing

sloppy thing to it then you're probably fine but if you're doing this and you find that through iterations it kind of bounces up and then bounces way back down and then bounces up and it looks really kind of inconsistent does have this nice kind of slope to convergence that's usually a telltale sign that you're running into some local minima or you know something's going wrong um in which case you might want to like tweak your priors and see if it works better or you might that might tell you there's just something wrong with your model um so it's a good kind of diagnostic i i just would love to hear like sometimes it's hard to grasp how can it be converging downhill like you like when we were talking earlier about the difference between sampling based methods and the factorized methods so there's obviously a world of difference in the setup and in the computation but like how is it that just doing something like minimizing free energy does take us to these acceptable parameter ranges if there's any sense that you have of working so closely with it um like what is it that that's what is it about that accumulating bars that leads us you know inevitably seemingly at least

reasonably to an acceptable set of
parameters in a vast
space what is actually happening there
that somehow it doesn't converge into a
local minima
or you know which way you go it doesn't
converge in a local solution set
oh there's no there's no guarantee that
it doesn't do that
usually this could easily converge into
a local minimum that's not the global
minimum
um you know i mean all this is literally
all this is saying is that there's a
kind of
smooth you know like locally convex
right like gradient descent towards some
stable value
um but no i mean that could totally be a
local minimum there's nothing that that
stops it from being that it just means
that there's a kind of
clear slope it's not like a super kind
of like bumpy
weird you know parameter landscape but
then that we need to inject some layer
of sampling first with many starting
positions
in order to to get good meta conversions
because
it's not enough to look at one model
trace
and say like yep well this one converged
that model converged somewhere
and it settled in given a very complex
free energy
function but then we need to pull back
another level

right to get traces from many points in the landscape

um well i mean that's just a that's just another approach that is a potential you know like not you know like trying to

not testing out a bunch of different prior values and seeing if everything converges from a bunch of different prior values to the same spot

um like yes it is a limitation because this could represent a local minimum but if it if it um stably converges

on um like different reliable you know parameter estimates for each person

you know which which corresponds to a range of places in a landscape

then um then like those will be stable individual differences you'll probably get slightly different values if you start your priors at a different place

almost inevitably because the complexity cost will be less if you don't have to move as far

to find accurate parameters but so long as you're getting interesting variability and convergence for each person

has a nice kind of slope like this then um it's not like everybody

is is getting stuck in some particular um you know single

well or attractor of some kind um

but again the question isn't necessarily what the true one is per se it's

you know like can you find estimates

that you know provide
interesting individual differences given
a set of priors
um you know i mean it's it's it's
important to realize too and we
mentioned
mentioned this in the paper that um you
know typically
the model the model that will win the
model though have the best parameter
estimates
will typically be simpler than whatever
the true underlying generative process
was in the person that generated the
data because typically there are simpler
explanations
um than the ones that generate the ones
that actually generate
data in a complicated system like the
brain
you know so so you're not necessarily
treating this as though it's
the true one but if it gives you nice
convergence for each person
and you're getting nice individual
differences then it still is a it still
is a reasonable measure
of you know of individual of interesting
individual differences
um and if you can simultaneously do
things like
show that parameters parameter estimates
correlate with other things like um
so just you know i happen to have it up
um
you know so this is this you know eeg
ceg study that uh i'm uh in the process
of putting a paper together before

and we have parameter estimates here um
for intraceptive precision um in that
gastrointestinal task
task i mentioned now precision here
correlates with reaction times
negative point point negative 0.74
despite the fact that the model is not
fit to reaction times
um and it's a direct prediction of the
neural process theory
that as interrupts you know higher
interceptive precision
should mean faster evidence accumulation
which should mean
faster convergence time which should
mean shorter reaction times
right so you know if you have you know
so you can separately validate that your
parameter estimates are tracking what
you want them to be
by seeing what they correlate with um
you know
in a way that they ought to correlate
with for construct validity for things
that didn't get fed into the model
um you know so there's there's lots of
ways of doing this um
um but but you're right i mean if you
using other approaches that use multiple
seeds the kind of like the monte carlo
stuff that you were talking about before
that is another approach and um you
might be more likely to
um you know find like a global
minimum um but uh but again they're just
they're just different approaches
there's just a quick plug so i think the
model that's kind of most closely

related to active inference and is
actually fittable
fitted can be fitted to data is probably
the hierarchical gaussian filter
um and i think they're actually gen at
the moment
it's estimated the parameters of the
rest made they're using variational base
but as i understand that they're also i
don't know if they have developed it or
are still developing
but they're also developing monte carlo
markov chain version
um very exciting yeah this is all just
yeah so so okay so last thing here
just uh just to um to you know to kind
of
you know wrap up everything so at least
people know
what um know what's available you know
for them to kind of expand on this in
our example code that we've
um you know set up is um
so down here uh when we actually do the
um where is this
uh so this this right here
is the function for doing model
comparison
so spm underscore bms bayesian model
selection
you just feed in the free energies for
each person
so for instance if you um
you know if i just uh
if i so i just it will just store these
right so for instance
four um so these
are the negative free energies for

um my six simulated people for their
best fit models
um best fit parameters uh for the two
parameter model
whereas that's it for the three
parameter model
and um and so
bayesian model comparison is just
um just feeding you just feeding both of
those vectors
into column vectors into that function
and it will spit out the expect expected
or predicted exceedance probability for
you as well as a number of other sort of
diagnostic checks and you can um
read uh there's good papers um like
class stuff on
and um anyway we we reference them and
they we cite them in the paper
that describe this in detail if you go
into
the actual uh you know function itself
it also
will tell you exactly what the inputs
are and exactly what the outputs
mean right so alpha is just a vector of
the model probabilities
again x_p is the exceedance probabilities
before doing the
protected part and so forth
so once that's all done
and i should say i'm not going through
every kind of line in this but
we commented it you know as well as we
could and we hope it's
we hope it's clear as you go through you
know what we're doing um but it also
will spit out

right cleanly into the into you know the
main
uh terminal here what uh you know what
the outputs are
um so if you wanna you can kind of
hopefully adapt
this code for your own purposes to do
recoverability and
um do model estimation and stuff like
that um
but once you've done that um
then you can do i was just going to show
you briefly
the um the hierarchical base the
peb um stuff and for that we put things
into
these gcm structures so gcm for model 2
gcm for model 3
and you can just pick which one
here that you want that you want to use
right whether you want to do peb on the
two parameter model outputs or the three
parameter model outputs
um and this just sets a bunch of
defaults in peb you shouldn't have to
mess with any of those
but if you but if you run it so i've so
what i've done is in the glm and the
general linear model for peb
i've just set the mean value across
subjects i've put a
uh regressor for group so this is saying
compare groups
um and then i've put in this sort of
fake age variable
just with random numbers basically so
if i run this section on the three
parameter model

then i'll get you know it'll do this
kind of estimation thing
and it will spit out a bunch of things
um and basically the way
to read this is um
ignore that so this one just ignore
uh when it spits it out this one will
tell you
um for instance this is saying like four
uh so this one is probably the best one
um so these are the reduced models
so the best model um removing the
parameters
uh parameter differences that um didn't
matter didn't win in bayesian model
comparison um so this is saying
parameter two and parameter three
were different between groups
and were and remain different between
groups and the best fit model
and again we we explained this all in
more detail in the paper
you can really ignore these ones at the
bottom
in the in the figure uh in the paper we
show you
which ones are um which ones matter
to look at such as these more or less
those top three
so the you know the estimated
differences and the estimated
differences that survive model
comparison
so kind of pruning away the parameters
that don't stick around in the best fit
model
um so in this one this is just gonna say
what the pink bars are

because to me it wasn't immediately that
they're the they're just the variances
they're just the variances and the
posterior estimates um
um so so uh so yeah so i mean for
instance
so you'd expect right this one has a
really high variance you you might
expect this one wouldn't survive
um you know as a winner in the in the
reduced model um
but even even this isn't probably the
most
useful the most useful um is the actual
like peb review parameter
gui um see so this is the actual
glm so this is just saying the mean the
group difference so this grape light
gray versus dark gray because the first
three subjects are a group
second three are a group this third
column is just the randomly generated
age values
um so you can just say for instance
you know which of these which of the
means are different well the means for
one and two are different
um you know so there's a main effects
essentially of those two
um you could go to group and
you could say okay these are the two
that are different
but you can put a threshold on it right
so weak evidence you know posterior
probability has to be above
you know some ever some value for
positive evidence
right so if i make it if i give it a

higher it has to there has to be decent
evidence for it then only the second
parameter here is the one that actually
has good evidence for a group difference
um you know for example so you can
you know but if i make it required to
have very strong evidence
so posterior probability greater than
0.99 then the whole thing gets removed
um so so this is a nice and you can you
know just do it in terms of free energy
or just in terms of
probability that a parameter is greater
than zero
you know so there's different ways to uh
different ways to do it but so this is
actually probably the nicest thing for
navigating
the the results of peb that will matter
if you're doing group analyses using it
um so
um so that um so that is
pebb um and um yeah i'm trying to think
if there's
anything um anything else really
um because that's i mean that's
primarily where where things end so i
mean at that point you should be able to
um should be able to do
everything um i guess the
yeah i mean i was gonna briefly i did
skip over this just the
this was the thing i mentioned that
philip um did where he was able to
do kind of like within trial by trial um
model predictions as opposed to these
just group differences and show the
trial by trial the

beta updates um correlated with this
midbrain
dopamine area but um but yeah i mean
really i mean that's more or less um the
steps you need to know
right so so what we talked about was you
have to have participants
perform the task you build one or more
models we did two
find the parameter values in each model
to best reproduce uh the data
right the behavior we did that we did
model comparison
we used the correlations between true
and generative parameter are
generated in generative and estimated
parameters
to make sure the winning model was
identifiable
and then you could do either normal
group level frequency statistics or
something like pep
to test for a between group or just
uh yeah group level um
you know between subjects sorts of
analyses on parameters at the end of the
day
um so i think i think that um
yeah i think that really covers um most
of the
you know kind of meat of it i mean the
only oh okay
sorry one other thing that i do need to
show you um
is that the um to actually
do the parameter estimation um
that calls where is it um
that calls this estimate parameter

script which is one that we also included

so if you open that just right click it and say open

um then this is where you set your prior means and variances

so here so it says here we specify prior expectations for parameter means and variances right so we say that um

so this is where so in this case i just set the same prior variance for all the parameters you know one over four you don't have to do that but so here you can think about it as the smaller the value you put here the greater the complexity penalty that you're adding right so the smaller that is the more you're going to prevent overfitting um but if you make it too tight then then the posterior estimates probably won't be that accurate

so so in this case what you can see that we've done is

um so if you're estimating alpha so action precision

then this is where you'd put the prior so we put a prior here of 16

um we log it here

i'm just so it's in log space um and that just makes it so during estimation um it doesn't allow that number to ever become

a negative value um so it keeps it it needs to stay

a positive value during estimation if if estimation ever tried to test out a negative value for that parameter then the thing would break

um same thing if if we were doing beta
so that would be
um which we didn't try here but i just
like we include it if you want
to so this would be the policy expected
free energy precision
so there the prior is one and again
we're just keeping the prior variance
um equal to one-fourth you know up there
for everybody
this is the loss of version which we
didn't include
um but is in there if you want to that's
the risk seeking parameter and we gave
it a prior of five
and then eta the learning rate this has
to be
not just any positive value it has to be
a value between 0 and 1
which means that you have to put it into
logic space as opposed to log space
and that just means that to set the
prior you have to make that
whatever your prior is and then also
put that same number here um so in this
case we've
given it a prior for learning rate of
0.5 which is just kind of halfway
between zero and one so it's just kind
of middle of the road
you know which you know would make sense
in a lot of cases
um and then more or less this function
down here
just turns those back into their normal
values so it exponentiates
the um the log values
um you know and so forth um so it just

brings them back
re-transforms them um
and uh this is just where we
um for technical reasons this is where
we set what the uh
what the values are for risk seeking
um and uh and then this is the actual
log likelihood
so it will just for each trial it will
add the log likelihood so the
mdp.p here is the probability of each
action
um that it will show that just as in the
mdp structure
which again we showed last time in in
other
sessions and in the actual um we have a
table
in the um table three i think in the
tutorial that says what each mdp field
is so it just takes those and just adds
them just logs them and it just adds
them
up um across trials to get the total
probability
and then the the less negative this is
at the end the better
the the um the better the the better the
fit is
um so it's important to know about this
one just because this is where you would
change your prior means and your prior
variances um
so so um so yeah and then
and then uh this for instance this dcm
field thing it's just set up so that
you would just enter the name of
whichever parameters you want to

estimate for that model so here if you
put an alpha in rs it'll estimate
alpha and rs but you could also add you
know eta here
right if you wanted to you could add
that then it would also estimate
anyway i mean i'm sure there's other
little things in the in the code here
that could be explained in more detail
but
um you know this this hopefully will be
enough to to get
people going well thanks so much
ryan and christopher this was really an
awesome series it was our first
model stream and it was really just a
great learning experience for all of us
so i'm going to give some final comments
from the chat so
uh someone wrote thanks for the authors
continuous
updating of the paper it will be great
if these uh slides as
in their current version are somehow
made available at the end of the episode
so maybe like you mentioned
doing a supplemental file or something
for the paper
so that was one thing
for these just for these slides i like
to put them maybe so someone could say
oh that was slide 87
at hour three of this thing and then we
could just have this version
so that people could know where that
figure was because i know that they're
probably in other places or just copied
over but

that would be one is that
like i mean i guess i can imagine i
could imagine
putting together a powerpoint that just
kind of you know kind of concatenates
all of the slides that we've gone
through over these over these sessions
the vast majority of them will be
just the same as the figures in the
tutorial itself
but the best the best i could do
probably is to
put that as a supplemental file um in
the um
for the supplement or the supplementary
materials on the psy archive
uh version um that's probably that's
probably the only thing we could
imagine doing we could have a first um
page on the pdf or something that just
says here's a link to the these streams
as this version so that'll be one thing
and then a related question
was um just that future people wanted
future tutorials and model streams
so either both of you it's an open
invitation to just come on whenever the
time is right
and talk about any kind of modeling or
all these other cool ideas we've been
bringing up
so i mean i'm happy to continue this um
you know in a more kind of
yeah on the fly you know way as people
request things i mean but
but at this point i mean i would i would
probably need
um or we would probably need um any kind

of requests

right like what is it that people want

to cover because i mean now we've kind

of gone through

the tutorial start to finish at least in

broad strokes

um so i would need to know at that point

what

you know from now forward i need to know

what else people would want us to cover

because you know we've gone through most

of it

cool well this tutorial definitely took

us to kind of the brink

of all of these papers i maybe it'd be

interesting to

walk through a paper like a tutorial of

the paper especially this interoceptive

one like

how would we adapt that for other do

kind of a walk through that'd be one

thought or we could go into the

formalism side we could invite a

colleague

or collaborator who came from a

different perspective more on the

analytical

more from some other dimension that

we're

that we are just also all learning about

yeah i mean i should say i mean there's

certainly a lot of areas of like the

broader kind of free energy literature

free energy principle literature that

we're

totally not covering here intentionally

right like our focus is this is what you

need to know

to build models and use them
experimentally
right so you know for instance there's a
lot of other stuff in
for instance like the physics
formulation for
for you know that also gets called right
part of active inference
um or the free energy principle more
broadly that includes things like for
instance like
um carl fristan's kind of variant on
talking about markov blankets
um you know or i mean there's a lot of
stuff on them like uh non-equilibrium
steady states
right and how that how that can be
talked about in terms of um
uh minimizing variational free energy
but also a lot of the physics
um you know like thermodynamic right
like free energy and things like that
also come in and um
that stuff's quite a bit more
theoretical and um um
it's certainly interesting but it's not
uh
not things that you need to know or all
that directly relevant to actually
build models and do the stuff in
practice so um
yeah there's other places other places
to look to that can they get that kind
of stuff i guess this is what i'm
pointing out as ours
this is not true completely
comprehensive of look it's called the
active inference or at least free energy

principle there sure
so any other final thoughts from either
of you
no i'm just been really fun thank you
yay great well it was really fun same
yeah and like i said i mean if if people
want to like
you know like go through get walked
through specific
you know papers either like you know
simulation either some of like our
simulation papers or empirical papers in
more detail
um so they could get a sense of how they
would do that in practice i'm happy to
do that
um but um i guess that's something that
we can just kind of
uh determine at a later date nice so we
will
wait for somebody to stimulate
us to do a walkthrough on a certain
topic or on a certain
um with a certain distribution of people
we'll figure it out but we'll um
we'll figure it out then so really
thanks again both
i'm gonna finish the live stream so
peace out everyone thanks for watching
activeinference.org
thanks so much ryan smith christopher
white